

ECE 315: Signals and Systems

Homework 8

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April 20, 2026

This homework is due on **4/27, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts: Fourier transform, convergence conditions) Consider the following continuous-time signals:

- $x(t) = e^{-|t-1|}$;
- $x(t) = \delta(t+1) - \delta(t-1)$;
- $x(t) = e^{-2|t|} \sin(t)$;
- $x(t) = \begin{cases} 1-t^2 & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$

Compute the Fourier transform $X(jw)$ for each continuous-time signal. Determine which type of convergence holds for the Fourier representation.

2. (2pts: Inverse Fourier transform) Consider the following Fourier transforms:

- $X(jw) = \frac{\sin(2(w-\pi))}{w-\pi}$;
- $X(jw) = \cos(2w + \frac{\pi}{3})$;
- $X(jw) = 2(\delta(w-1) - \delta(w+1)) + 4(\delta(w-\pi) - \delta(w+\pi))$;
- $X(jw) = \begin{cases} 1 & |w| \leq 2 \\ 0 & \text{otherwise.} \end{cases}$

Compute the continuous-time signal $x(t)$ for each Fourier transform.

3. (1.5pts: Fourier transform properties) Consider the following continuous-time signal $x(t)$ with Fourier transform $X(jw)$:

$$x(t) = \begin{cases} 2-t & 0 \leq t \leq 1 \\ t & 1 \leq t \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Without explicitly computing $X(jw)$, apply the Fourier transform properties to compute the following quantities:

- (i) The phase of $X(jw)$;
 - (ii) $X(j0)$;
 - (iii) $\int_{-\infty}^{\infty} X(jw)dw$;
 - (iv) $\int_{-\infty}^{\infty} |X(jw)|^2 dw$.
4. (1pt: Fourier transform, convolution) Consider the following continuous-time linear-time invariant systems with impulse response $h(t)$ and input $x(t)$.
- (i) $h(t) = e^{-t}u(t)$, $x(t) = te^{-t}u(t)$;
 - (iii) $h(t) = \frac{\sin(4(t-1))}{\pi(t-1)}$, $x(t) = \cos(6t + \frac{\pi}{2})$;

Compute the system responses for each case using Fourier transforms of $h(t)$ and $x(t)$.

5. (2pts: LTI systems by differential equations) Consider the following continuous-time stable and causal linear-time invariant system with input $x(t)$ and output $y(t)$ that is described by a differential equation:

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 2x(t).$$

- (i) Determine the impulse response $h(t)$ of this system;
 - (ii) Compute the output of the LTI system for the input $x(t) = te^{-2t}u(t)$.
6. (1.5pts: LTI systems by differential equations) Consider the following continuous-time linear-time invariant system with input $x(t)$ and output $y(t)$:

$$x(t) = (e^{-t} + e^{-2t})u(t) \text{ and } y(t) = (2e^{-t} - 2e^{-3t})u(t)$$

- (i) Determine the frequency response $H(jw)$ of this system;
 - (ii) Determine the impulse response $h(t)$ of this system;
 - (iii) Find the differential equation relating the input and the output of this system.
7. Read Section **7.9**, and complete the problem **7.9-4** in **LSS**.